This implies a completely automated unmanned production unit, to be supervised by humans over computers only.

Automated machineries are making production processes error-free and increasing production volumes. For example, FAD 2500, a fully-automated dispenser by Musashi Engineering comes with an automatic dispensing volume management system to ensure consistency of dispensing volume.

Robotic technologies are being introduced in industrial machines to optimise their performance. Kenmec Mechanical Engineering demonstrated the inclusion of floor pattern sensors in its industrial vehicles. These can speed up material carriage and logistics while reducing mishaps of vehicles, like disbalance on rough floors, falls from elevated areas or staircases, and so on.

Industrial power management has also improved significantly. We witnessed a power conditioning system by Delta Electronics, which claims to have 97.6 per cent efficiency. It can be used for applications like power quality improvement, load shifting, peak shaving and so on. Users also receive energy analytics for reference.

Indian intelligent robotics firm GreyOrange demonstrated its smart linear sorter and intelligent pick-and-place robot, Butler, which can massively improve warehousing and reduce operations time.

Smart IPBX solutions like Crux LX by Singapore-based Crux Labs and Moreconn, by Taiwan-based Mobiletron, demonstrate how geographically-distributed units and offices can stay connected through audio as well as video, at zero cost.

Popularity of KNX protocol is growing considerably as firms like Mean Well, Schneider and ABB, among many others, are producing KNX-compatible components to make industrial and building automation simple. KNX-compatible products can synchronise with each other to create a complete automation ecosystem that can be utilised in applications like lighting, industrial power failsafe units, medical electronics and more.

Test equipment are adopting the low-power wireless approach, as integration of NB-IoT, Wi-Fi and Bluetooth support are increasing exponentially.

Meter Industrial showcased a group of air-quality and water-quality monitors that are NB-IoT- and Bluetooth-enabled, and send readings wirelessly in real time.

We came across a multichannel LoRa tester by Good Will Instruments, which integrates all major LoRa tests and claims to save testing expenses by up to 70 per cent.

Consumer electronics

Our homes have become a hub of smart devices, essentially making our lives easier and more comfortable. While smart speakers like Amazon Echo or Google Home, and smart lights like Philips Hue, often come up as some of the most popular home automation gadgets, we came across some interesting solutions.

> As we go ahead, we foresee innovations dominated by robotics, AI, the IoT and renewables

Muse, a smart band to be worn across the forehead, provides EEG results on its accompanying mobile application. It senses the level of calmness of the user and provides real-time tips for relaxation.

For controlling household appliances, we found i-Control Wi-Fi Remote Control Box by AIFA Technology. It converts a smartphone into a remote for any IR-powered appliance (TV, air-conditioner, set-top box, remote-controlled fan, light, etc) through 360-degree IR control.

Micrometre-scale LEDs can redefine TV design and use. Samsung showcased the first modular microLED 371cm (146-inch) TV, called The Wall, at CES 2018. It is a self-emitting TV with micro-LEDs that eliminate the need for colour filters or backlight while producing superior viewing experience.

Bone-conduction technology is an upcoming trend in earphones. In this technology, sound is transmitted across the ear bone instead of occupying the ear canal, enabling users to listen to audio and even speak while keeping the ears free to hear external sounds. Players like BOCO and Human are developing such solutions.

Smart speakers remain popular gadgets, as the year started off with Lenovo announcing its smart speakers in the market.

Computers are also expected to become faster and more powerful, as this year witnessed the launch of Intel i9 processors, which have an overclocking ability of up to 5GHz.

Other applications

There are many more interesting use cases of smart technologies. For instance, we came across a robot that is being used by banks and police for assistive operations.

There are examples of 2.5D printers, brought by players like Casio, that use near-IR lights to generate slightly-elevated print patterns.

Many innovators are working on flexible and foldable digital displays. These are estimated to become a trend in smartphones in the foreseeable future.

Self-driven cars are already being researched, as many technology firms are working on designing secure and precise advanced driver-assistance system (ADAS) platforms.

Drones are becoming increasingly popular as these are being designed for a range of use cases, starting from simple videography to carrying out precise farming in farmlands, as well as for surveillance using thermal imagery.

Way to go

As we go ahead, we foresee innovations dominated by robotics, AI, the IoT and renewables, as these technologies will drive applications in smart cities, buildings, factories, agricultural lands, healthcare and more.

Law authorities are deploying robotic workforces to improve legal compliance. Doctors are referring to AI platforms to ensure 100 per cent accurate diagnoses and treatments.

Corporates are investing in renewable energy sources to reduce environmental impact.

In these ways, we will see more interesting things happening going ahead, and we will look forward to searching them out! EFY